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Horizontal Well Acid/Propped Fracturing and Production Optimization via Multiple Hydraulic Jetted Laterals

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Abstract

Successful Transverse fracturing from horizontal wells can suffer several undesirable behaviours. During the pumping phase, point-source growth behaviour and stress-state driven geometry can be problematic, and then under production, radial convergence and pressure/conductivity deterioration with distance from the wellbore. While Longitudinal fracturing corrects these issues, it suffers from a complete failure to project the drawdown laterally away from the wellbore. However, today the ideal scenario can be readily achieved through the use of optimum Longitudinal fracturing of Transverse jetted lateral-pairs from conventionally drilled horizontal wells.

Transverse fracturing is the dominant horizontal well stimulation geometry; but suffers a number of well documented shortcomings whereas the advantages of Longitudinal fracturing are themselves offset by poor drainage. A new technique of mechanically jetting opposing pair(s) of laterals, 600 ft tip to tip; followed by hydraulic fracturing combines the best of both of these geometries. This technique, when applied to a conventional horizontal well drilled in the minimum stress direction, offers multiple advantages. These advantages are related to improved hydraulic fracturing properties (reduced pressures, direct geometry control and enhanced fracture conductivity); as well as direct production enhancement, (complete linear inflow behaviour, projection of drawdown and improved drainage, EUR and RF over similar flow periods) when compared to conventional Transverse fracture placement.

Lateral jetting, in vertical wells, is well understood; and has been a standard intervention practice over many decades. Thus, the placement of opposing laterals within a horizontal is merely an extension through a mechanical hurdle. Industry experience has demonstrated that such laterals can repeatedly penetrate as much as 100 m from the wellbore. Subsequent acid/propped fracturing of lateral-pairs results in numerous advantages. Emanating from a line-source, instead of a point-source, elliptical fracture growth results in surgical in-zone placement, with less dependence on the stress state. Initiation pressure and transient net-pressures are reduced due to the flow geometry. For both gel and acid, the ability to rapidly transport fresh/raw fluid to the lateral tip completely changes effective conductivities and acid-effectiveness, resulting in increased and enhanced width and conductivity.

However, it is under production that the dramatically impacted aspects can be appreciated via this fracture placement approach. The drawdown itself is projected, by the presence of extreme or infinite conductivity tunnels, much closer to the tip of the fractures, thus resulting in an increased effective drainage radius of the overall system. Inflow to the fracture from the reservoir is initially linear, then Linear again from the frac to the lateral itself and then finally linear to the wellbore also. Pressure drop through the conductive fracture is minimised due to the distribution of flow. The result is improved Productivity Index, drainage and EUR. Multiplication of this effect, using optimised pairs of laterals along a conventional horizontal well delivers considerably improved economics, with reduced fracture and well count as a byproduct.

This new technique (referenced herein as the "CTS" system/approach/method)for stimulating tight reservoirs allows for enhanced recovery, reduced well / frac count, optimised capitalisation and improved economics. This technique offers a revolutionary option, as well as a means of developing reserves that are currently stranded due to their: poor fraccability, challenging reservoir geometry, non-uniform drainage or undesirable contacts. Similar to the combination of fracturing with horizontal wells, this enhanced projection of the drawdown into formations along with fracture geometry control will create many new development opportunities particularly in conventional but tight-reservoirs.

Introduction

Multi-stage Transverse hydraulic fracturing (Figure 1) operations from horizontal wells (both acid-fracturing and propped-fracturing) presents us with a potentially undesirable conundrum. The geometry of such fracturing, under a production scenario, achieves the very desirable effect of projecting the drawdown deeper into the reservoir, albeit through a decaying fracture conductivity and quality with distance from the wellbore. However, the initiation of the fracture as a point-source at the wellbore, can create a constraint in terms of fracture initiation/breakdown pressure, fracture half-length (x_f) to fracture height (h_f) aspect ratio, perforation-friction and also Near-Well-Bore Friction (NWBF) loss. Under production, while the drawdown is projected, the radial convergence at the wellbore can also result in a Non-Darcy (ND) skin effect, particularly in the presence of multiple phases. One such case is with Retrograde Gas-Condensate (RGC) reservoirs, where the CTS approach not only assist with flow regimes and Non-Darcy effects, but has the added advantage of projecting draw-down (and therefore the dew-point location) deep into the reservoir remediating condensate-banking.

Conversely, multi-stage Longitudinal fracturing (Figure 1) from horizontal wells delivers unparalleled ease of initiation, alignment of the fracture with the wellbore and an unzipping of the perforated interval over an extensive length, assisted throughout by the Kirsch effect (Kirsch, 1898). Furthermore, the resulting initiation from a line-source geometry means that an extensive fracture half-length (x_f) to fracture height (h_f) aspect ratio can be achieved along the lateral length. However, while a Longitudinal fracture provides a perfect route for Linear flow into the wellbore the immediate projection of the drawdown is only across the vertical height, and is not extensively projected into the reservoir, severely limiting the effective drainage area.

It is very clear therefore that in an ideal World, we would be able to combine the positive characteristics of Transverse and Longitudinal fracturing from the fracture initiation and the fracture growth process as well as the subsequent fracture productivity and fracture drainage considerations. Additionally, in a similar fashion, if combined effectively we would completely eliminate any of the negative consequences of each of these different geometric completion approaches. The remainder of this paper will demonstrate how this combination can now be readily achieved, and how examples have already been executed in both vertical (Potapenko *et al.*, 2021) and horizontal wells (Rylance & Korovaychuk, 2022) in various parts of the World.

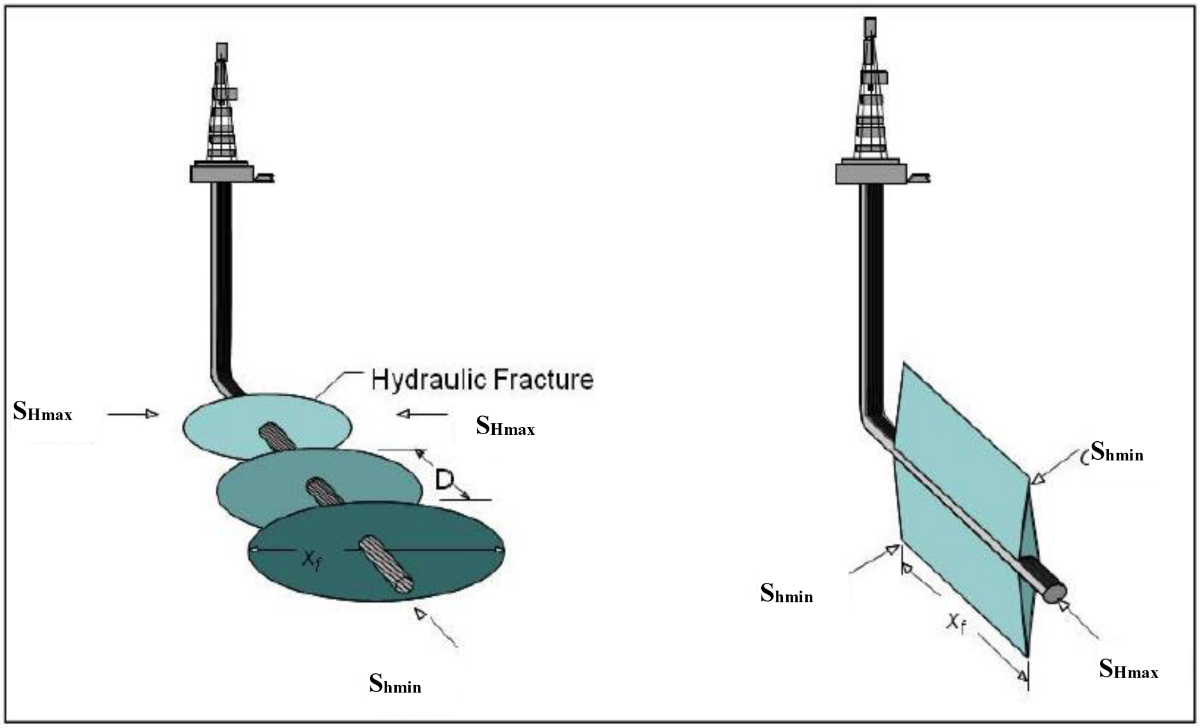


Figure 1—Transverse Fractures and Longitudinal Fractures after (Economides & Martin, 2010).

This novel technique for efficiently stimulating tight reservoirs allows for the delivery of more effective, more efficient and enhanced recovery, using reduced well and fracture counts, with optimised application of capital and improved economics. In a very similar fashion to the combination of hydraulic fracturing with horizontal wells, this new enhanced combination of fracture and wellbore geometry creates a truly 3D hydrocarbon offtake geometry.

Transverse Fracturing

Transverse fracturing has increased in popularity since the onset of horizontal well drilling and hydraulic fracturing have been combined in conventional Oil & Gas fields; and particularly since the extensive development of unconventional resources. For the reasons noted above, Transverse hydraulic fracturing has proven to be the most effective in terms of individual well EUR as well as drainage footprint below a certain mobility threshold (Liu *et al.*, 2013). However, in order to fully appreciate the nuances of these aspects we will consider the two areas of the Fracture Placement and the Fracture Productivity of Transverse fractures separately.

Transverse Fracture Placement. For Transverse hydraulic fracturing from a horizontal wellbore, then from an end view we can see the potential progression of the created fracture geometry in an unencumbered stress-state, in a very simplistic cartoon fashion in (Figure 2 Left). For Transverse hydraulic fracturing the hydraulic fracture initiation, in both cased-cemented and open-hole environments, has been shown to potentially be complicated by the creation of a Longitudinal fracture component at the wellbore (Figure 2 Right). This near wellbore complication is primarily due to the Kirsch effect (Kirsch, 1898), which is discussed later. While this effect is often a major disadvantage to Transverse fracturing, it is considered a major bonus for Longitudinal fracturing.

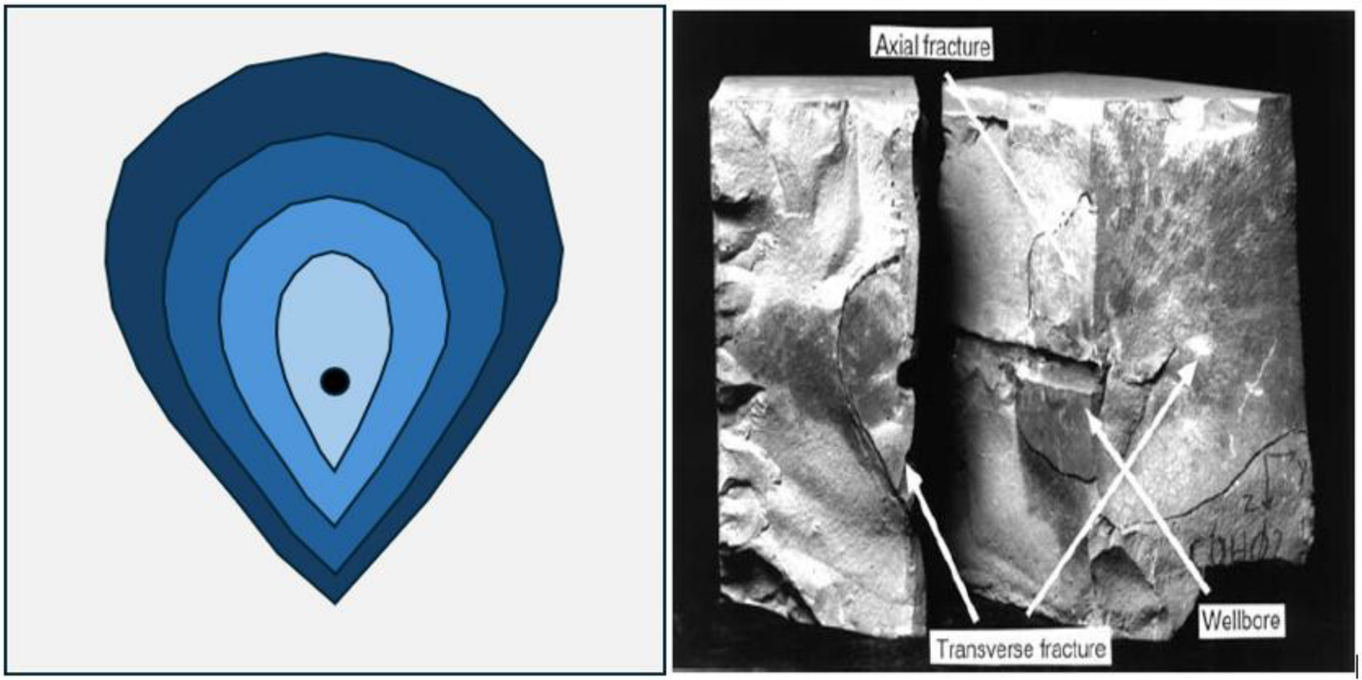


Figure 2—Radial to Elliptical Point-Source Fracture Growth (Left) and Longitudinal Component of Transverse Fracture. (Weijers *et al.*, 1994).

As can be seen in (Figure 2 Left), typically within a uniform stress-state, and in the absence of extensive stress-barriers, a differential between the minimum in-situ stress (σ_{hmin}) and the fracturing fluid density (ρ_f), creates fracture growth that moves rapidly from an initial radial state to an elliptical/upward growth bias. Achieving any kind of fracture geometry aspect ratio (h_f/x_f) is extremely difficult, and even in the presence of modest barriers, extensive fracture half-length is only realistically created at the expense of excessive fracture height. This effect dramatically impacts the fracture coverage efficiency and also increases the required well-density (reduces spacing). Hence, within conventional (tight-rock) hydraulic fracturing the way that the geometry is presented to the reservoir is absolutely critical, as fracture spacing (usually single-digit) and well counts are generally more limited.

With conventional hydraulic fracturing in tighter-rock typically being performed in hard-rock environments, frequently utilizing viscous gels and quality propping material, often the NWBF from the Longitudinal fracture component (as seen in Figure 2 Right), can result in significant issues that may need to be addressed. The pressure-loss, and width reduction often associated with subsequent fracture realignment, can result in severe bridging occurring and near well Screen-Out (SO). Additionally, proppant-pack pinch-points resulting from this fracture transition zone and post pump-shutdown slurry movement, can result in ineffective connectivity between the principal fracture body and the wellbore.

Transverse Fracture Productivity. When considering the productivity of Transverse fractures, we can see from (Figure 3 Left) that, until intra-fracture interference takes place and subsequently pseudo-radial flow, that we have an efficient form of Linear flow into the fracture-face. The flow is then produced at and into the wellbore via a less efficient converging Radial flow regime, (see Figure 3 Right). In the presence of more than one producing phase within the fracture, then significant pressure loss may result (Soliman *et al.*, 1990). This is particularly true under initial production conditions, i.e. during the clean-up and subsequently as the individual fracture production rates are at their highest.

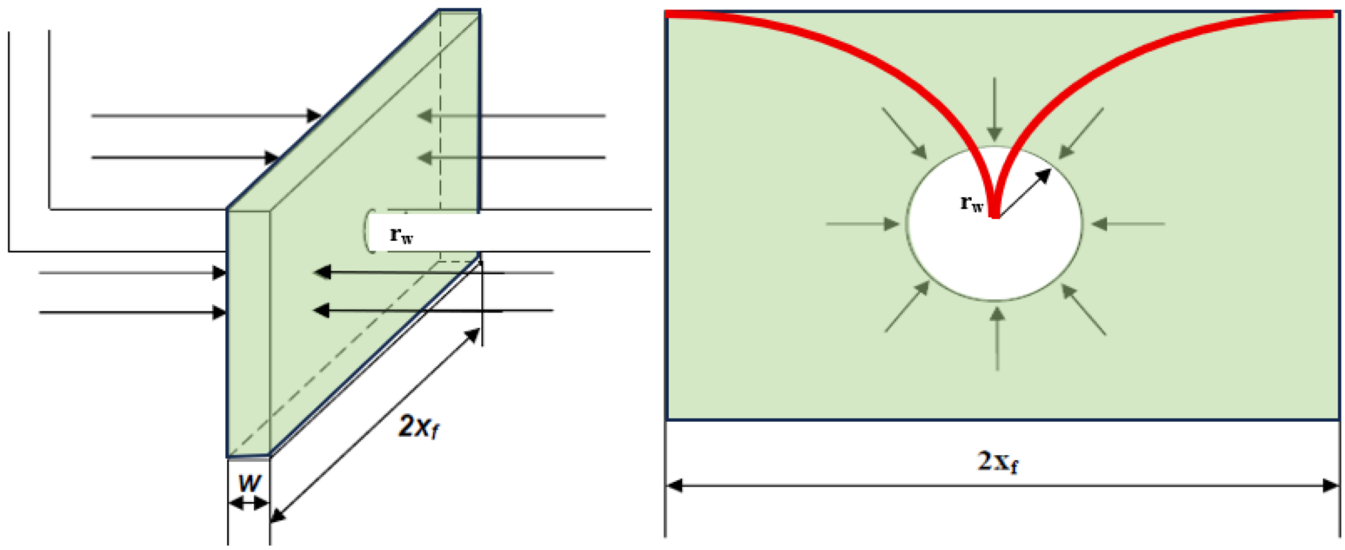


Figure 3—Linear Flow into Fracture Face. (Economides & Martin, 2010) and Radial Convergent Flow within Fracture into Wellbore as well as the Lateral Deterioration of the Drawdown (in RED).

We can also see (from Figure 3 Right) that the projection of the effective drawdown along the created fracture half-length x_f (noted in red) continually deteriorates with distance from the wellbore, due to a combination of multiple factors. These include:

i. Convergent flow at the wellbore;
ii. Multi-phase fluid environments, including condensate-banking in RGC systems,
iii. Deterioration of fracture width w_f with increasing distance from well and hence effective fracture conductivity $k_f w_f$,
iv. Deterioration of effective fracture conductivity with time, particularly in unpropped acid-fracture treatments; and also,
v. Long-term deterioration of the fracture conductivity for a variety of other reasons (Vincent, 2009).

The result is that while the fracture wellbore geometry results in the creation of a 3-Dimensional offtake system; that the action of drawdown is fully effective in one of these dimensions (mother wellbore length), and in the case of the other two dimensions (fracture height h_f and fracture half-length x_f) then only partially effective.

Longitudinal Fracturing

A number of broad industry studies and investigations have been performed (Wei & Economides, 2005), (Economides & Martin, 2010), (Liu *et al.*, 2013) and (Yang *et al.*, 2015); to demonstrate that the economic transition region (between the selection and application of Transverse and Longitudinal fracturing) is principally based upon the mobility ratio (k/μ). It is not appropriate to debate the reasoning for that in this paper, but it is sufficient to say the economic transition point at which it is better to have Longitudinal fracturing is not as high as most people would expect to observe. Typically, one may conclude that the Industry' applications of Longitudinal hydraulic fracturing geometry in a general sense, have been in a much more bespoke fashion than that of Transverse fracture geometry. Some examples of the application of Longitudinal fracturing include (Coghlan & Holland, 2009), (Allan *et al.*, 2010), and (Nagar *et al.*, 2016) and it may be noted that these selections have principally been driven by very specific considerations.

These considerations can include:

i. Plans for future Enhanced Oil recovery (EOR) via producer well conversion and subsequent waterflood, where a desire to create a 'wall-of-water' and avoid short-circuit(s) in the system is required.
ii. The anticipation of extensive production related, near-wellbore Non-Darcy pressure losses, and their calculated impact on the production rate are simply too impactful (particularly within high-rate, multi-phase environments) and
iii. Finally, large reservoir aspect-ratio(s) (length-to-width) in reservoir geometry and orientation with regard to the principal in-situ stress-state; might require that fracture/wellbore geometries be placed specifically to simply minimize the number of required well penetrations (and drill centres, platforms and manifolds) to develop a particular field.

The conditions, as noted above, hint at the complexity of the selection process (and the underlying conundrum of selection) which is worth further discussion. Therefore, in order to fully appreciate the nuances of these aspects then we will next consider the two areas of Longitudinal fracture placement and the Longitudinal fracture productivity of Longitudinal fractures separately below.

Longitudinal Fracture Placement. For Longitudinal hydraulic fracturing, placed from an appropriately oriented horizontal wellbore drilled in the σ_{Hmax} direction, then considering a simple 2-D well (side-view) we can see the progression of a created fracture geometry initiated within an unencumbered stress-state, shown in (Figure 4) below.

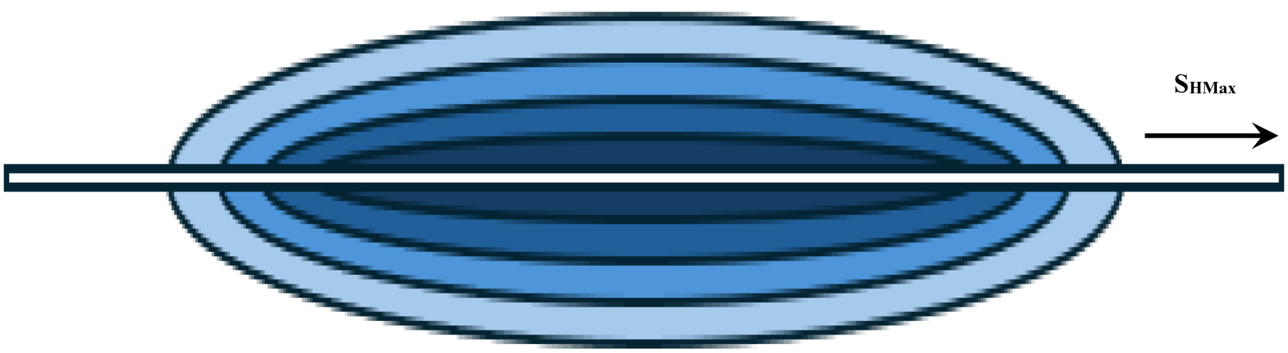


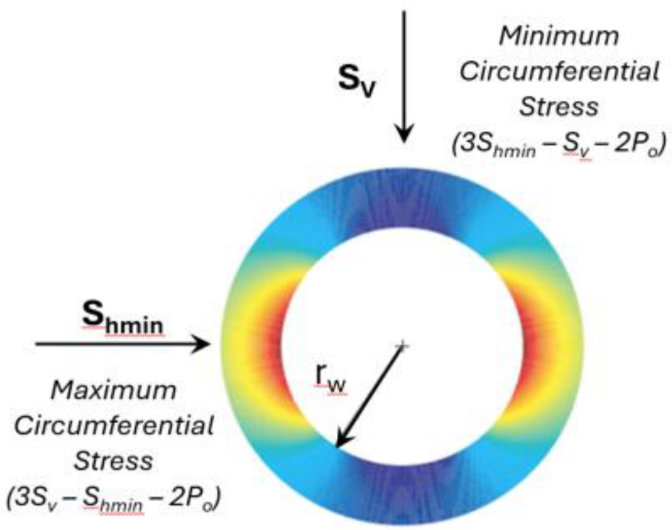
Figure 4—Elliptical Line-Source Longitudinal Fracture Growth.

As can be seen in (Figure 4), assuming both a uniform stress-state and the absence of extensive stress-barriers, then the fracture growth would initiate elliptically (Martins & Harper, 1985), initially unzipping along the wellbore. Continued pumping produces further elliptical growth and yields the essentially confocal elliptical patterns as depicted in (Figure 4); i.e. a pattern that sequentially approaches an ever more radial geometry. While the in-situ fracture gradient (and gravity/density) creates a similar slight upward bias (in the fracture height growth), an effect exaggerated in Figure 2 Left, the advantage given by this elliptical growth pattern is due to the fact that the fracture is emanating from a line-source. Compared to the fracture growth from a conventional set of perforations or indeed a Transverse orientation, wherein the fracture fluid advances and radial fracture growth occurs from essentially a point-source, the action of a largely continuous plane of pressurized fracturing fluid acting from a horizontal borehole wall is a very different scenario. Such an approach is fundamentally both mechanically and geometrically different. In addition, of course, the fact that, in both Normal and Strike-Slip stress regimes, this horizontal lateral wall has an initial tensile-force top and bottom, as defined by the Kirsch effect (a more detailed discussion to follow later). *In summary, a more favourable setting for creating a vertical, Longitudinal, controlled fracture height geometry can hardly be imagined*. The result being, especially in conventional, tighter rock formations, that this entire

Longitudinal fracture is developed almost contiguously and simultaneously along a significant portion of its entire tip to tip length ($2x_f$).

Accordingly, the fact that this elliptical fracture geometry is developing from a line-source; wherein the fracture aspect ratio (h_f/x_f) is extremely constrained, then dramatically limits the effect of any vertical height growth, let alone a bias. Thus, achieving the desired outcome of a significant fracture geometry aspect ratio (h_f/x_f) through constraining the created fracture-height (h_f) becomes a relatively straightforward exercise; one that begins with a confocal ellipse along the wellbore. These very positive outcomes have been observed many times and in multiple settings, where Longitudinal fracturing has been rigorously applied.

As alluded to earlier and noted in (Barree & Miskimins, 2015), for Longitudinal hydraulic fracturing, both the hydraulic fracture initiation and growth processes are significantly assisted by the Kirsch effect (Kirsch, 1898), see Figure 5. This figure describes the particular case of the hoop stresses ($\sigma_{\theta\theta}$) around a horizontal wellbore, drilled in the direction of S_{Hmax}. In fact, if the classification of the in-situ stress regime is either Normal or Strike-Slip, then for both cases the vertical stress will be the Maximum of the two stresses acting around the wellbore and the Kirsch-defined placement of the opposing pairs of tensile and relative magnitudes of the compressive forces about the borehole circumference will be the same. Thus, in both of these stress-states the Kirsch effect will dominate, as seen in Figure 5. Note that the unique case of a reverse stress-state, wherein any fracturing operation performed would invariably lead to a horizontal fracture system, is considered an unnecessary diversion for this discussion.



We can describe the wellbore wall stresses as follows.

$$\sigma_{\theta\theta min} = 3S_{hmin} - S_v - 2P_o$$

Where:

$\sigma_{\theta\theta min}$ = Effective hoop stress around wellbore,
S_{hmin} = Minimum in-situ stress (horizontal),
S_v = Vertical Stress (typically overburden),
P_o = Wellbore (internal) pressure.

Figure 5—Hoop Stress Distribution for Longitudinal Case.

Therefore, for a wellbore drilled in the direction of S_{Hmax} as the internal wellbore pressure is gradually increased, during the initial wellbore breakdown process, a Tensile stress develops at both the 0° and 180° Degree locations around the wellbore wall; as depicted by the dark-blue region in Figure 5. This tensile region invariably results in the fracture initiating and 'unzipping' along the top and the bottom of the open-hole lateral length. While this behaviour is extremely undesirable in Transverse fracturing leading to tortuosity and associated issues, see Figure 2 Right, in Longitudinal fracturing it is highly desirable and results in an incredibly effective fracture/wellbore communication at the initiation stage.

Furthermore, this Kirsch behaviour provides for a significant reduction in the required Fracture Breakdown Pressure (FBP), which has already been reduced due to the jetting effect removing inherent tensile strength. In tighter conventional fracturing environments this can make all the difference between fracture initiation failure or success; or at a very minimum remove the requirement for excessive and costly additional intervention work such as abrasive-jetting. Note that application of excessive breakdown pressures can also negatively impact stage to stage isolation, either internally due to plug failure and also externally due to packer, swell element or cement channelling failure.

Longitudinal Fracture Productivity. When considering the productivity of Longitudinal fractures, we can see from (Figure 6 Left) that, throughout the production phase including pseudo-radial flow, that we have an efficient form of Linear flow into the fracture-face. The flow is then produced at and into the wellbore via a similar and highly efficient Linear flow-regime. In the presence of more than one producing phase within the fracture, then significant pressure losses are now minimised by more evenly distributing the flow into the wellbore.

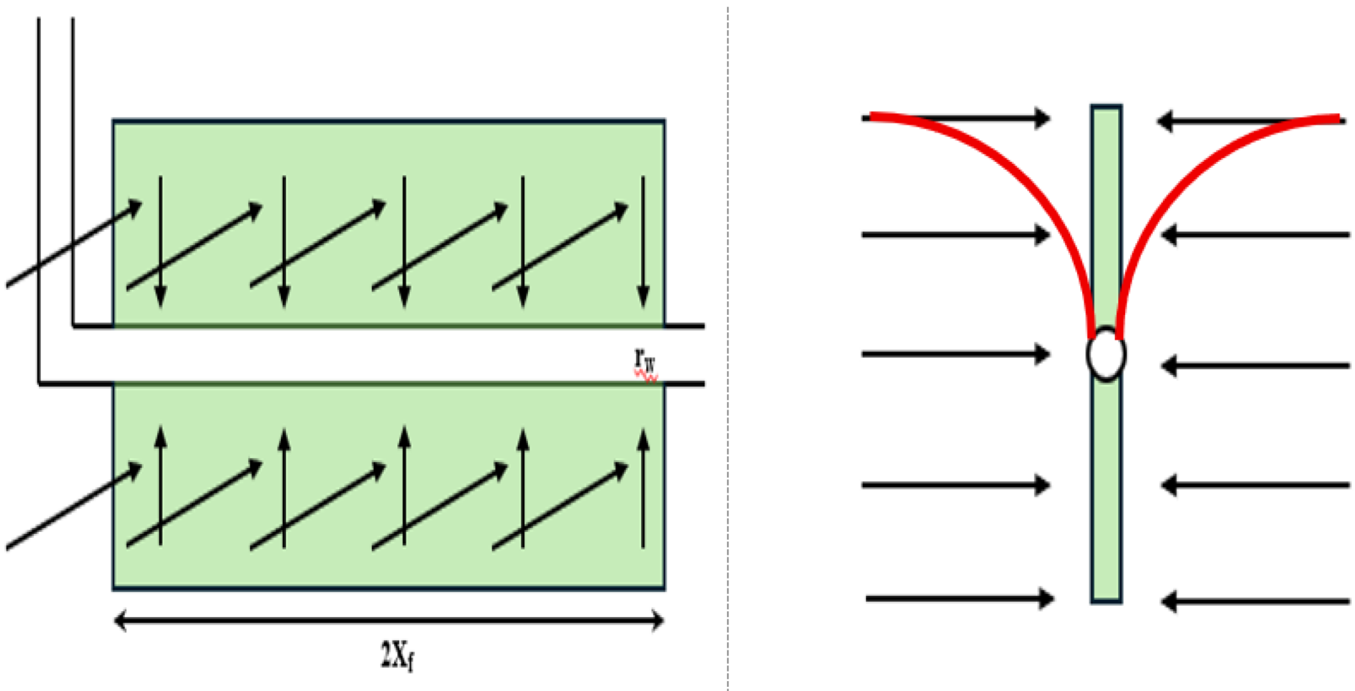


Figure 6—Linear Flow into the Fracture Face (and Wellbore) (Left) and Drawdown through Formation (not the Frac) (Right).

We can also see (from Figure 6 Right) that the projection of the effective drawdown is through the formation (noted in red), along the fracture half-length itself, the drawdown is considerably higher due principally to the immediate and lengthy connectivity with the wellbore. Furthermore, as the fracture half-height ($h_f/2$) is considerably less than the (x_f value), and the flow-rates are reduced to the Longitudinally distributed flow, then the deterioration of the effective $k_f w_f$ over the fracture half-height is a fraction of that experienced with the Transverse fracturing approach.

The result is that while the fracture wellbore geometry only results in the creation of a 2-Dimensional offtake system; that the action of the drawdown is fully effective within both of these dimensions, namely the mother wellbore length (fracture-half-length x_f), and also over the fracture height h_f. This is an extremely effective offtake, but as noted unfortunately fails to project the drawdown out into the reservoir (e.g., outside of the common plane of the fracture and wellbore) in a 3-Dimensional manner.

Idealised Fracturing Case

It can therefore be surmised, that if we could affordably make each Transverse fracture (in a multi-stage fractured horizontal wellbore), look and behave like Longitudinal fractures, that we would create the ideal 3-Dimensional offtake system. Under this scenario the fracture initiation would be a low FBP, with similarly reduced fracture propagation pressures yielding a line-source generated elliptical geometry, resulting in a desirable and dominant aspect ratio (h_f/x_f). Then under production, inflow would be Linear into the fracture face, as well as Linear into the wellbore, with shortest flow-path(s) possible through the fracture conductivity and an associated minimisation of total pressure losses within the system.

The remainder of this paper will now clearly demonstrate that such an ideal Transverse/Longitudinal fracture geometry creation and production system approach is now entirely feasible to create, simply by combining/extending a suite of existing technologies in such a way as to achieve the best of both worlds. The various component parts, influencing factors and methodologies will be explained and described; as well as areas of further investigation which will offer even more potential value. Mathematically sound, with a robust engineering basis this novel approach will result in enhanced economics, opportunity and applicability of multi-stage horizontal well hydraulic fracturing operations; beginning with those environments where current practice is sub-optimal or has not yet been able to overcome the technical hurdles that exist. It is suggested that there are numerous conventional environments where matrix permeability continues to deliver the ability to drain, but where the inherent mobility requires significant and controlled surface area generation through hydraulic fracturing. In addition, that those environments would significantly benefit from this ideal Transverse/Longitudinal fracture geometry and that such an application could be the next step in economically unlocking significant trapped resource.

System Description

So, from the discussion(s) held within the previous Section, we can deduce that the desire is that we are able to create a highly efficient Longitudinal fracture by hydraulically fracturing a jetted lateral orthogonal-pair, which has been created Transversely from a horizontal mother-bore. Additionally, that this process can be effectively and efficiently repeated a number of times (typically < 10), at equal spacings along the mother-bore horizontal, see Figure 7.

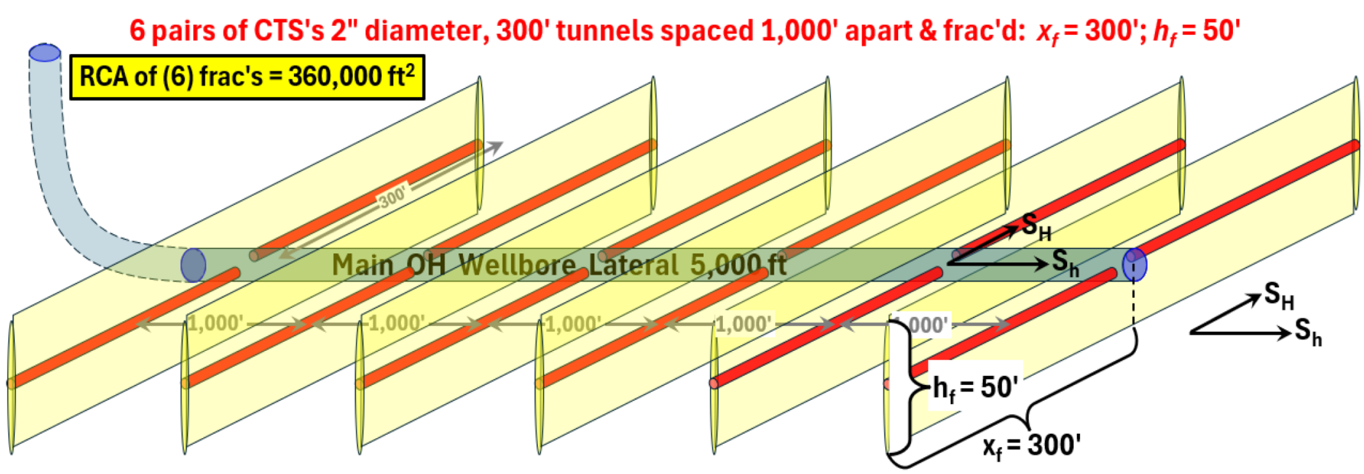


Figure 7—Example Wellbore with 6 lateral CTS lateral pairs each individually hydraulically fractured.

Ultimately, this is the lateral, fracture and wellbore combination that we wish to create; in order that we may effectively access all of the advantages of the fracture growth and production behaviours of both the Longitudinal and Transverse fracture approaches while at the same time avoiding all of the disadvantages. We will now discuss how this is achieved, how the laterals are physically constructed, how the fracture geometry develops and the resulting production from a suite of these systems within a well.

System Jetting

The purpose of this Section is to describe the manner in which the pre-fracture tunnel-pairs are formed, and more specifically, the downhole systems that have been designed in order to create these (roughly

2″ diameter by 300 ft long) lateral boreholes. For the sake of simplicity/reference, this lateral jetting approach will simply be referred to as the CTS system; so as to distinguish it from a number of similar, but fundamentally differing approaches, which exist within the industry Today. The lateral creation systems, with which we will compare CTS, will be referred to here as (i) Abrasive Jet Perforating (AJP), (ii) Coiled-Tubing Drilling (CTD) and (iii) Radial Jet Drilling (RJD). While these systems are able to deliver some of the necessary characteristics of a Dual-Lateral pair, as described above, only the CTS approach can achieve the complete suite that is required and then repeat this for multiple sets within horizontal wells. This unique capability is demonstrated in Table 1 and also within Figure 7 above.

Table 1—Lateral Construction System Characteristics.

Characteristics	AJP	CTD	RJD	CTS
Creates laterals from highly deviated/horizontal wells.	YES	YES		YES
Creates laterals using and erosive high-pressure fluid-jet.	YES		YES	YES
Creates laterals using an abrasive high-pressure fluid-jet.	YES			YES
Creates laterals by constant advancement of BHA system.			YES	YES
Creates laterals using mud-based systems if required.		YES		YES
Creates laterals in Cased or Open-Hole conditions.	YES	YES		YES
Creates laterals that exit orthogonally from the wellbore	YES		YES	YES
Creates lateral pairs that can be up to 300 ft in length.			YES	YES
Creates laterals with BHA in Compression or Tension				YES

CTS and AJP Systems. AJP is really as stated, a simple perforating system (Brown & Roper, 1960); the jetting nozzle, is stationary laterally relative to the formation, is always contained within the mother-bore and creates penetration of just a few feet. Whereas with the CTS system, this is a tunnel-jetting system, which extends tunnels up to 300 ft from the mother-bore deep into the formation. However, both the AJP and the CTS systems are similar in that they both provide an orthogonal exit from the mother-bore. AJP can however tunnel into a formation sufficiently to provide fracturing fluid with the necessary connectivity with a formation to allow fracture initiation to take place (Guizada, *et al*., 2017) due to a reduced FBP. Therefore, the CTS jetting-system can effectively be described as taking the capability of the conventional AJP methodology and extending this into the formation some 300 ft orthogonally from the wellbore.

CTS and CTD Systems. CTD began in the Early 1990s (Wesson, 1992) and (Traonmilin et al., 1992) and has gone on to be a well proven technique for achieving enhanced Reservoir Contact Area (RCA) and increasing this value from the RCA of the main well mother-bore alone (Al-Omair *et al*., 2011; Alshuhail *et al.*, 2023 and Guizada *et al.*, 2018). On occasion CTD may be referred to as a fishbones drilling technique, not to be confused with FISHBONES completion technique (Al-Saedi & Dunbar, 2025 and Dehdouh *et al.*, June 2024). Although both techniques refer to those cases where multiple open-hole laterals are drilled as offshoots from the mother-bore creating a fishbone pattern; with FISHBONES system laterals being limited to an effective Range-2 (R2) joint-length due to the deployment technique. However, in all these cases such systems are referred to as being potentially in lieu of hydraulic fracturing, rather than being combined with it. This is the principal reason that such approaches have been limited in success in tight-rock and instead have been deployed in those areas where reservoir heterogeneity play a more significant role. In summary, when it comes to cost effectively creating RCA, it is very difficult for any alternative to compete with some form of hydraulic fracturing.

CTS and RJD Systems. Superficially, the CTS approach may initially look to be equivalent to the RJD technique (Dickinson & Dickinson, 1985), however upon closer examination and consideration of the characteristics (see Table 2) then we can see that they are in fact very different. Fundamentally, the CTS system, is considerably more robust in comparison to typical RJD jetting systems. The principal difference between the CTS and RJD systems is related to the jetting-hose, the jetting nozzle and the manner of deployment of the process.

Table 2—Comparison of RJD and CTS hose and systems.

Typical Characteristics	RJD System	CTS System	Units
Hose Geometry			
Hose Length	328.0	300.0	ft
Hose External Diameter	0.551	1.500	in
Hose Internal Diameter	0.375	1.000	in
Hose Cross-Sectional Flow-Area	0.098	0.590	in^2
Hose Operating Characteristics			
Hose Burst Pressure	5,802	24,000	Psi
Hose Minimum Bending Radius (MBR)	3.540	5.000	in
Hose Max. Pump Rate	15.100	84.000	gpm
System Operating Characteristics			
System Pump Pressure (minimum)	5,656	10,000	psi
System Pump Pressure (maximum)	7,107	15,000	psi
System Hose Pressure Loss	2,390	1,072	psi
System Nozzle Pressure Loss	3,237	2,500	psi
HHP @ Nozzle (Minimum)	49.900	490.080	HHP
HHP @ Nozzle (Maximum)	62.700	735.120	HHP

While the RJD system is run in tension (due to a largely flexible low-rigidity hose), the CTS system hose (see Figure 8) is designed to allow for deployment under either tension or compression. The preferred mode for the CTS jetting operation is in compression, under which the entirety of the pumped-energy available can be used for the jetting erosional effect; and with an extremely stiff hose this is readily achievable.

When jetting in tension, this can require that as much as 60% of the jetting energy has to be re-dedicated to rearward facing jets, in order to generate sufficient forward drawing force, note that this is the only methodology for the RJD system. With a burst pressure rating of 24,000 psi, the CTS system jetting hose (even with a 5″ bend radius) has an equivalent yield-strength similar to N80 pipe and is more than capable of being deployed in compressive mode.

The significantly increased ID and pressure-rating of the CTS jetting hose, compared to that of an RJD, is what enables the order-of-magnitude increase in the throughput of jetting fluid, and hence the drilling force (HHP) delivered to the jetting nozzle. In order to generate jetting pressure drops across a nozzle that exceeds a given rock's threshold failure pressure, the RJD nozzles rely on multiple, very tiny (often 1 mm or less) diameter jets, and therefore require a very clean, solids-free jetting fluid. As the CTS nozzle can handle 2 bpm or more and do not require rearward facing thrust jets, it can use a single forward jet of a diameter equal to those jets as used in AJP (1/8th to 1/4th inch) or even larger, such as those used in drill bits (8/32^{nds} to 10/32^{nds} inch). This is what enables the CTS system to utilize, if necessary, abrasive jetting slurries or even drilling mud, and still generate the necessary exit velocities.

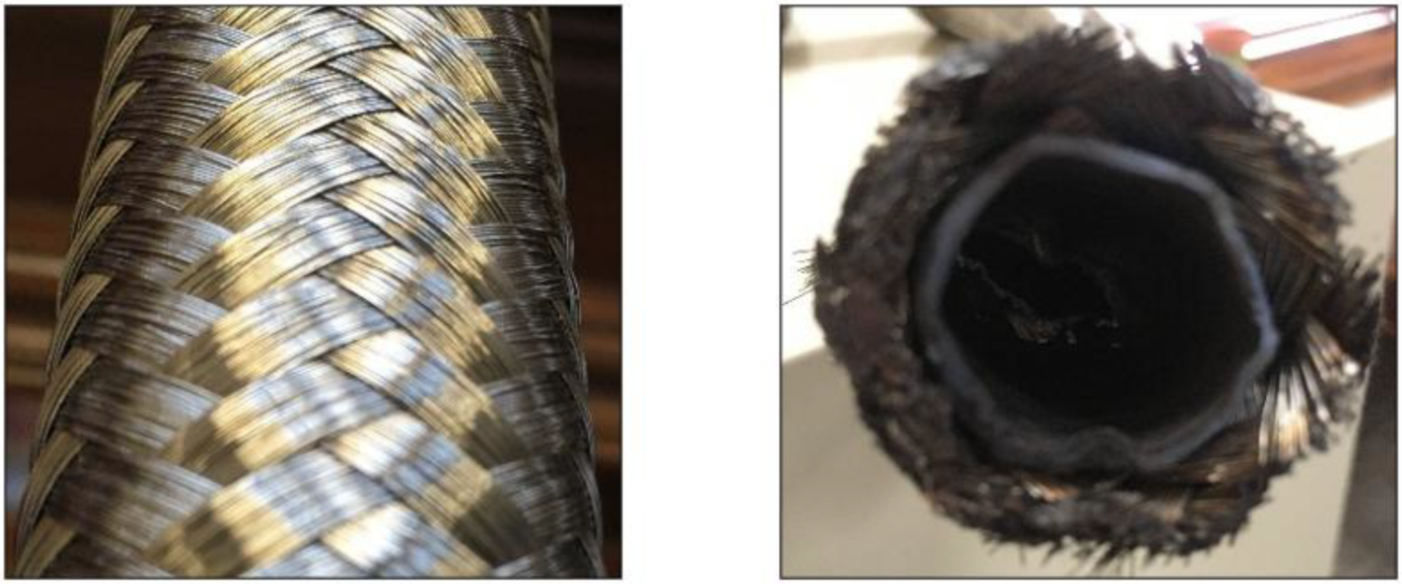


Figure 8—CTS Composite layered hose (before and after shearing).

System Fracturing

As noted in the earlier section on Transverse and Longitudinal fracturing, the fundamental reason for creating the opposed jetted laterals is so that we can fracture the lateral pair Longitudinally (while actually being geometrically Transverse from the mother-bore). The intent is that for each of the jetted-lateral pairs, e.g. six stages as noted in Figure 7, that we can individually place the hydraulic fractures, before flowing-back and commingling for the production phase. There are multiple ways that this particular phase could be achieved, with two distinct options that we will refer to as Pre-Liner and Post-Liner.

Pre-Liner: The first of these approaches would likely be under a simplistic barefoot scenario, whereby the drilling-rig would TD the horizontal mother-bore, at which point a CT unit would RIH, space-out and jet the opposing lateral pairs in a fully barefoot condition. Then upon completion of the requisite number of lateral-pairs, the drilling rig would then run the production-liner with a suite of standard sliding-sleeves, which would be spaced close to each of the lateral-pair depth(s) along with an effective external segment isolation method. This approach then leaves the well in a condition where each of the individual lateral pairs can be effectively hydraulically fractured (propped or acid) in isolation, sequentially from the toe to the heel of the horizontal well and then be commingled for flow.

Post-Liner: With this approach the drilling-rig would TD the horizontal mother-bore. After this the drilling rig would then run an uncemented production-liner, with a suite of specialised sliding-sleeves (one for each planned pair of jetted-laterals) as well as the necessary external casing isolation (either mechanical or swell element). At this point in time the drilling rig would RD and a CT unit would then be RU and would access the open-hole through the specialised sliding-sleeves, an also jet each lateral pair orthogonally at the same sleeve location. After each pair of laterals have been jetted, or alternatively the whole suite of lateral-pairs has been completed; then acid or propped hydraulic fracturing may be performed on each lateral pair through each of these specialised sliding-sleeves.

As noted in the earlier section on Longitudinal fracturing, the orientation, extent and open-hole nature of the jetted lateral result in a line-source fracture being generated from tip-to-tip of the jetted laterals. This fracture will grow out along the lateral-pair, from the centre source, with limited height growth, until the fracture reaches the lateral tips (see Figure 4). At this point in time, further fracture half-length (x_f) growth, at the extremes of the laterals, will take place; however, this additional half-length will be generated at the cost of more extensive fracture height (h_f). While lithology, in-situ, fracture gradient, rock-mechanical properties, fracturing fluid viscosity and pump-rate will all have influence on the fracture growth; these are secondary phase influences on the initial geometry generated which is completely dominated by the line-source (and Kirsch) behaviour.

It can be seen, that using this specialised geometric fracturing approach, that it is possible to generate extremely favourable aspect ratios of the created fracture half-length (x_f) to the fracture height (h_f). In addition, this approach also results in an underlying uniformity and regularity between the suite of fractures that are created; due to the dominant nature of the jetted-lateral pair to the fracture geometry creation; such repeatability is rarely available in the hydraulic fracturing arena. For any particular application, the landing point of the horizontal would be selected so that the created (height) coverage achieved was as desired. Note that creating fractures in this way allows them to be engineered (i.e. fracture coverage can be readily achieved) in high-stress formations, above or below lower stress regions, thief-zones or depleted layers; which would otherwise normally dominate geometry control.

For fracturing, the presence of a substantial 2″ diameter jetted lateral, placed some 300 ft either side of the mother bore has a number of additional positive influences on the fracture placement process and resulting stimulation effect. In the case of acid-fracturing operations, this channel will accelerate the raw acid to the tip of the fracture and significantly enhance the nature and evolvement of the temperature profile, acid spend and generation of substantial fracture-face worm-holing. Such projection of raw acid, across a line-source to the fracture tip, will require less retardation and result in enhanced stimulation distribution. In the case of propped hydraulic fracturing, a significant change in pressure distribution due to the presence of a line source will mean that although the resulting aspect ratio is very much PKN (Perkins & Kern, 1961) by nature, the fracture width relationships will be more aligned with that of a GDK (Geertsma & deKlerk, 1969) scenario, with a maximum w_f achieved along the jetted lateral.

In all cases the opportunities that are presented, due to the presence of this large diameter channel, offer substantial potential for the enhancement of chemical and fluid placement, along with the conductivity of the proppant pack in the case of their use with propped hydraulic fracturing operations. Acid and propped fracture chemistry, schedule and design will require reconsideration in order to take full advantage of the characteristics that this geometry offers. Initial indicators are that substantial cost efficiencies will be achieved while at the same time improving stimulation effect and substantially reducing placement risks.

System Production

For a simple production/productivity assessment, then we would like to be able to directly quantify the potential benefits of this improved fracture placement approach compared to conventional fracture placement. Therefore, we wish to directly compare conventional radial point-source growth behaviour with this new elliptical line-source growth delivery; and this is most easily achieved by comparison with a real-life example of the Khuff reservoir in the KSA. There is much that has been published on the Khuff fracturing performance (Mahbub *et al.*, 2012), (Rahim *et al.*, 2013) and (Hussein Al-Jubran *et al.*, 2010); and it has also been well established that achieving consistent successful acid/propped hydraulic fracturing within this formation can present significant challenges.

Reservoir Contact Area (Point Source vs Line-Source). The basis of the comparison therefore will be through the use of a general Reservoir Contact Area (RCA) assessment between the two differing fracture methodologies. Figure 9 below, is reproduced from Islam *et al.* (2015) and refers to the RCA generated in the multi-stage acid-fractured horizontal wells of a Khuff target. It can be shown that, for each fracture treatment added to the well that a total of 24,000 ft^2 of additional RCA is generated, and with a formation gross thickness of 50 ft that this is generated by the creation of an effective fracture half-length of 120 ft.

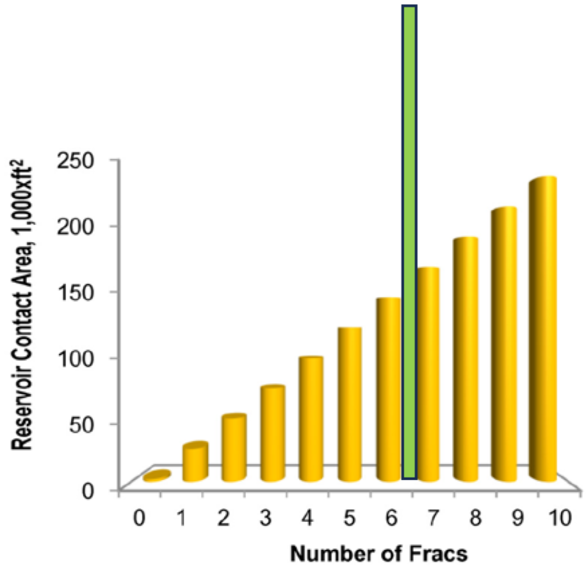


Figure 9—Increase of Effective Reservoir Contact with Frac Stage Count (after Islam *et al.*, 2015) and CTS approach.

The generation of this limited fracture half-length, in the conventional Transverse fracturing manner, can be seen with reference to Figure 10. This figure demonstrates the underlying inefficiency of point-source fracturing in an unhelpful stress-state; where the creation of the fracture half-length reaches a critical or 'asymptotic' value, past which point further half-length is difficult to achieve effectively and most certainly not economically.

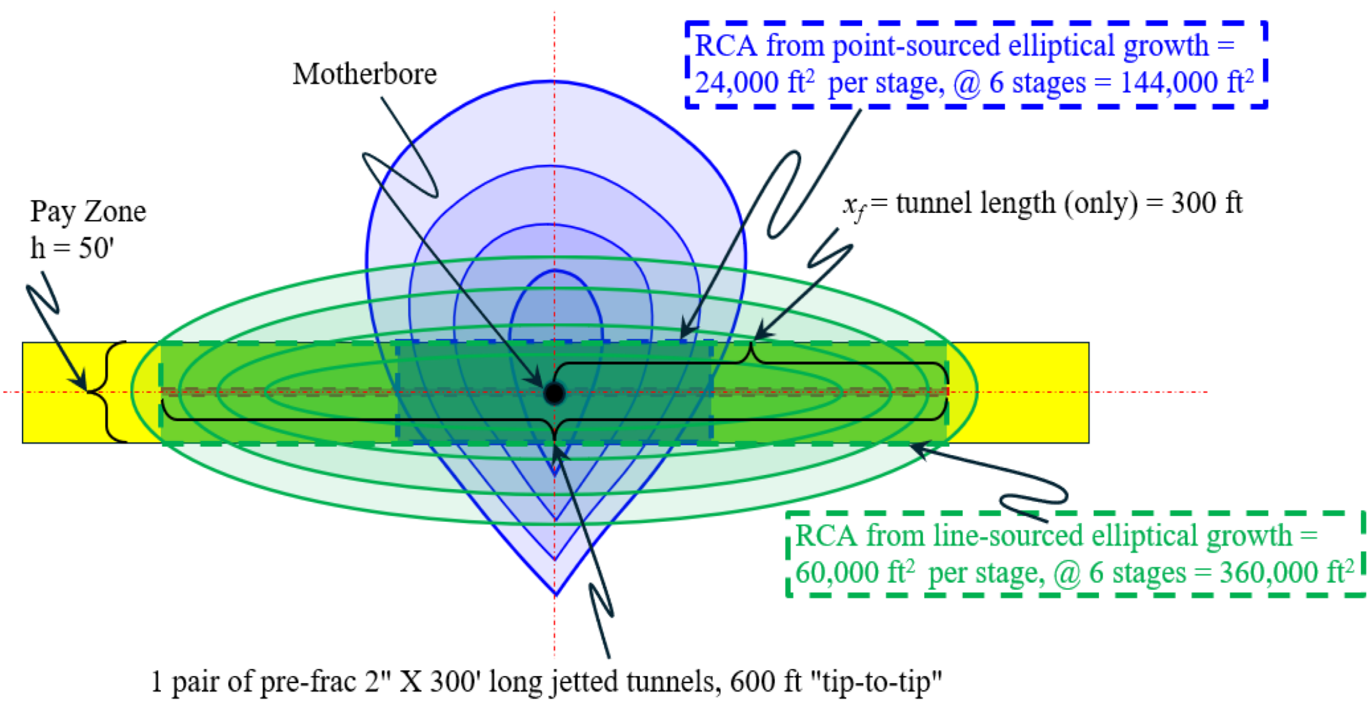


Figure 10—Create fracture half-length and RCA from Point-Source and Line-Source approaches.

Conversely for the elliptical fracture-growth case, from an initial line-source, then we can start with a profile view depicting the previously placed CTS-generated 2″ diameter -by- 300 ft long tunnels. Therefore, the x_f for such a line-sourced elliptical growth would assume, conservatively, that the line-sourced fracture half-length will initially grow no further than the pre-existing jetted channels, i.e., x_f = 300 ft, as depicted in Figure 10. This results in an increase in the RCA from 24,000 ft² to 60,000 ft², per fracture treatment, before taking into account any further fracture growth past the tip of the lateral.

A preliminary fracture model assessment was made by NSI, Inc.; that considered this approach in a generic high compressive strength, HPHT Middle East carbonate formation, similar to the Khuff. This work suggested that an additional achievable x_f growth of 100 ft was readily achievable, beyond the tunnel-tips. Note this is quite similar in magnitude to the x_f = 120 ft value obtained from simple point-source radial growth from the mother-bore. This extra 100 ft per wing would in fact, increase the baseline RCA from 360,000 ft² to 480,000 ft² in the hypothetical 5,000 ft lateral, 6 frac stage system.

Ahmed, M., *et al.* (2012) published a highly useful multi-stage fractured Khuff *Q-vs-t* type-well, which we can now use as a starting point, see Figure 11. This Khuff type-well is projected to have an EUR of 16 BCF over an economic life (t_{EL}) of 10 years. Note the precipitous drop in the early life production, from an IP (Q_i) of 20 mm.scf/d down to 8.35 mm.scf/d in just the first 100 days.

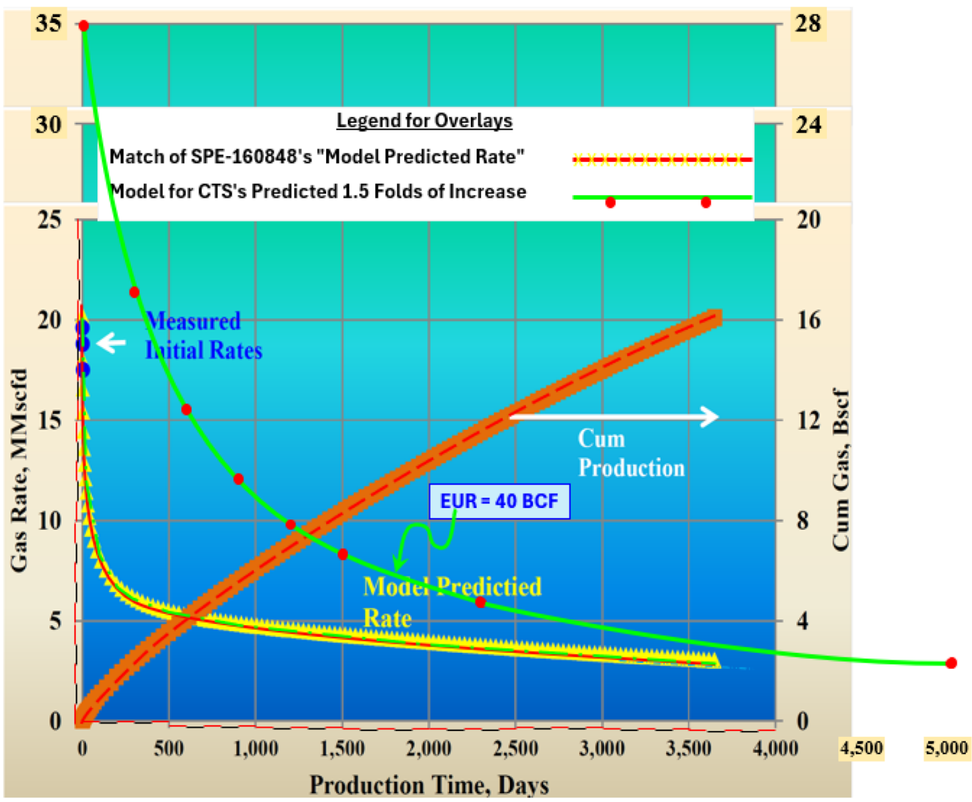


Figure 11—Long Term Production Forecast from (Ahmed, M. *et al.*, 2012)

As a basic approach we can simply adjust this simple baseline production curve, in direct relation to the response to RCA uplift and verify with some additional simulation. This results in an estimated EUR of 40 BCF, for a similar well lifespan of 10 years, although the flowrates remain reasonable at 2.9 mm.scf/d at the end of this period. Initial rates are of course substantially higher and sustained, and this capability has been demonstrated in other Khuff publications (Alabbad *et al.*, 2011); however, this approach would need to be combined with a re-design of the Upper completion based on the new performance envelope and longevity.

Reservoir Jetted Lateral Impact. It is important to note that this tunnel/fracture technology and their permanent nature, provides an enhanced and extensive $k_f w_f$, due to the presence of the 2" jetted channels themselves. Furthermore, the presence of these channels also offers the opportunity to both create and sustain the fracture conductivity (particularly with acid-fracturing) much more effectively, than would normally be the case under conventional approaches.

Fracture Creation: As described earlier, in the case of performing the acid-fracturing operations, the presence of this channel will provide an opportunity to accelerate the raw acid to the confocal outer boundaries of the fracture and significantly enhance the nature and the evolution of the temperature profile, the acid spend, and the generation of substantial fracture-face worm-holing. Such a projection of raw acid, across a line-source to the fracture tip, will require less retardation and result in an overall enhanced stimulation and fracture-face worm-holing effect and distribution.

Fracture Production: For any horizontal wells drilled in the direction of the minimum horizontal stress (i.e., Transverse fractures) the jetted pilot laterals can result in providing a major improvement in the post-fracture productivity behaviour. This is primarily due to the fact that the jetted pilot laterals would mitigate (eliminate) the converging flow effect at the wellbore, as described above. This could be even more advantageous for the particular application of acid fracturing. In that specific case, the high drawdown that occurs near the wellbore, and the resulting increases in the effective stress on the fracture, will further degrade the etched effective fracture conductivity in this critical converging flow environment. This drawdown-based flow restriction is now almost completely eliminated as the jetted pilot laterals carry the flow directly into the wellbore.

The existence of the jetted pilot laterals acts now to completely change the flow regime behaviour within the fracture. The traditional methodology of assessing a fracture effectiveness is with the use of the F_{CD} or Dimensionless Fracture Conductivity (FCD) approach (Cinco-Ley & Samaniego, 1980).

$$F_{CD} = \left(\frac{k_f w_f}{k X_f}\right) \tag{1}$$

This F_{CD} relationship describes how effectively the propped fracture plane transmits the flow from the fracture tip to the wellbore, i.e., the along the created fracture half-length (x_f). Desirable values of F_{CD} range from 2 for reasonably moderate permeability formations to perhaps as high as 20 for very low permeability formations, this for the case of vertical wells where the fracture fully communicates with the wellbore. However, mitigating increasingly converging flow with proximity to a horizontal wellbore requires much higher conductivity values, at least in the very near wellbore region. Under the CTS jetted laterals case however, the facture only needs to transmit the flow up or down to the pilot hole itself ($h_f/2$) – and not the entire length of the fracture (x_f) to the wellbore. For lower permeability formations where the fracture length is large and the height is controlled – this is a much shorter distance. In the case of the example that we have been considering this is 25 ft instead of 300 ft. Thus, the x_f term in the F_{CD} equation would be replaced with $h_f/2$ where h is the fracture-height/pay-zone thickness (assuming the pilot hole is placed in the middle of the pay). With this substitution, one might refer to this as F_{CD}-V (as in Vertical). Achieving a conductivity for an F_{CD}-V system of 2 is obviously much more easily created than the traditional fracture conductivity approach. Once again from our own example that has been used throughout, as the ratio of x_f to h_f is 16, then we would require 16 times more $k_f w_f$ to achieve an F_{CD} of 2 using the conventional approach than we would with the CTS laterals. Thus, the very presence of the pilot jetted laterals makes a significant contribution to post-frac production just by virtue of their enduring existence; and this in addition to the contribution achieved from improving the underlying fracture geometry.

Conclusions

The completion techniques of both Transverse and Longitudinal hydraulic fracturing each have their own very distinct advantages and disadvantages, and these have been outlined, described and referenced within this paper in some detail. What this new novel CTS (fracture geometry and fracture productivity/drainage) approach clearly achieves, is combining all of the advantages of the Transverse and Longitudinal hydraulic fracturing approach and eliminating all of the disadvantages of the Transverse and Longitudinal hydraulic fracturing approach. This unique completion combination unlocks, for the very first time, significant advantage in the area of fracture geometry creation and hydrocarbon production and areal drainage.

Fracture Creation: Longitudinal hydraulic fracturing of the 2″ jetted lateral tunnels then results in the creation of a geometrically controlled Transverse hydraulic fracture, with a substantially enhanced fracture aspect ratio (x_f/h_f). The fracture initiation pressure, the net-pressure trend and resulting fracture face worm-holing and proppant placement stimulation effect all benefit substantially from the presence of the jetted laterals. The velocity, temperature and distribution of the acid-chemistry/fluid-volumes that are injected and similarly the nature, pace and location of surface area created; are all positively influenced such that the resulting stimulation effect is enhanced.

Fracture Production: During the production phase, this enhanced fracture geometry, i.e. the presence of the jetted laterals and the considerably enhanced aspect ratio has a significant impact on the potential production rate. Improved fracture/reservoir coverage and massively increased x_f enhance the underlying productivity in a conventional manner. The flow regime(s) are now largely linear in nature, the distance of hydrocarbon travelling through conductive fracture body is $h_f/2$ and not $x_f/2$, and the conductivity will already have benefitted from the enhanced acid transport and placement. Finally, the projection of drawdown along the infinite conductivity jetted laterals (effectively a 600 ft smallbore horizontal well) substantially changes the flow-regime(s) during the drainage process.

Recommendations

This revolutionary new method of combining Transverse hydraulic fracturing with horizontal wells offers an opportunity to dramatically change the economics of developing tight reserves, where the conventional Transverse fracturing approach is currently being used and also when estimated to be inefficient, ineffective and uneconomic. For these reasons, in the same manner as when we combined hydraulic fracturing with horizontal wells, Operators should consider/re-consider those scenarios where conventional Transverse fracturing assessment is being applied or has been considered and rejected as being uneconomic. There are many scenarios, formations, reservoir geometries and in-situ conditions and characteristics where the use of conventional Transverse fracturing completion techniques are challenging, for a variety of differing reasons. The CTS approach combines the known benefits of Longitudinal Fracturing with the known benefits of Transverse fracturing and then enhancing those results through the presence of an infinite-conductivity stable channel. This approach offers Operators a new, novel and unique methodology to either dramatically enhance the economic value of existing Developments or allow Development of previously considered stranded resources.

Nomenclature

k = *Formation permeability (mD)*
ρ_f = *Fracturing fluid-density (lbs/ft³)*
μ = *Fluid viscosity (cP)*
$\sigma_{\theta\theta min}$ = *Minimum hoop-stress (psi)*
S_{hmin} = *Minimum in-situ stress (psi)*
S_{hmax} = *Maximum in-situ stress (psi)*
S_v = *Vertical stress (psi)*
h = *Reservoir height (ft)*
h_f = *Fracture height (ft)*
$k_f w_f$ = *Fracture conductivity (mD.ft)*
P_o = *Internal wellbore pressure (psi)*
Q_i = *Initial gas production rate (mm.scf/d)*
t_{EL} = *Economic Life of Well (days)*

w_f = *Fracture width (in)*
x_f = *Fracture half-length (ft)*
AJP = *Abrasive Jet Perforating*
CTD = *Coiled Tubing Drilling*
CT = *Coiled Tubing*
EOR = *Enhanced Oil Recovery*
EUR = *Effective Ultimate Recovery (mm.scf or k.bbls)*
FBP = *Fracture Breakdown Pressure (psi)*
FCD = *Dimensionless Fracture Conductivity (dimensionless)*
GDK = *Geertsma De klerk Khristianovich*
HHP = *Hydraulic Horse-Power (HHP)*
HPHT = *High-Pressure High Temperature*
IP = *Initial Production rate (mm.scf/d)*
KSA = *Kingdom of Saudi Arabia*
NWBF = *Near Well-Bore Friction (psi)*
PKN = *Perkins-Kern Nordgren*
RD = *Rig Down*
RGC = *Retrograde Gas-Condensate*
RIH = *Run In Hole*
RJD = *Radial Jet Drilling*
SO = *Screen-Out*
WO = *Work-Over*

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